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Title:

Lab 5 Architectural Desing

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BodyBalance AI: Requirements Report

CS3151 Software Engineering, Fall 2026

Abstract

BodyBalance AI is a proposed mobile health, fitness and recovery decision-support platform. It addresses a gap between goal-based fitness applications and specialized clinical services: active people cannot easily judge what activity is appropriate on a given day when they are in pain, tired, stressed or poorly rested. The system combines a daily physical and mental check-in, a structured Pain Locator, optional BodyScope AI posture analysis and personal history to recommend one of six outcomes, from maintaining activity to seeking professional evaluation. Predefined red-flag symptoms stop routine guidance and trigger referral. This report presents the analysis phase: the problem, a market and feasibility study, a business model, three personas, elicitation methods, functional and non-functional requirements revised to be SMART and testable, use cases, a context diagram, a communication view, a literature-informed system architecture and a project plan, together with alignment to the UN Sustainable Development Goals and Saudi Vision 2030.

Introduction

People who exercise regularly often know their goals but not what is appropriate today. Pain, fatigue, poor sleep or incomplete recovery change the right decision, yet most applications plan from long-term goals or track metrics without asking how the person feels. At the other end, rehabilitation services give clinical care but are not built for daily training choices. BodyBalance AI is intended to sit between the two by answering one question: considering how I feel and what is happening with my body today, what is an appropriate next step?

This report presents the analysis phase of BodyBalance AI, a health, fitness and recovery decision-support platform. It gives the project requirements in their final form for the midterm discussion: the problem, the market, the business model, the users, the requirements and how they were gathered, the use cases, the system boundary and the plan for managing the project.

Scope and limits. The first release covers a focused minimum viable product. Specialized scenarios such as combat sports, older adults, first aid and sensitive injury documentation are future work. Numeric targets and the red-flag list are initial values that need validation with users and healthcare advisers.

1. Problem to be Solved

Active people often know their fitness goal but cannot tell what is appropriate on a given day. Poor sleep, stress, fatigue, recurring pain or incomplete recovery are rarely considered by general fitness applications, which plan around long-term goals.

The resulting situation has four parts:

- Condition-related information is scattered across separate applications, notes and memory.
- A photo or a single symptom report shows only part of the picture.
- Users cannot tell when pain is routine soreness and when it needs professional attention.
- Coaches and clinicians receive incomplete, unorganized histories.

Root cause. The information needed to judge a person's condition today (pain, sleep, stress, recent activity, history and images) is held in different tools or never recorded, and no tool turns it into a decision with a safety check.

Why an automated solution is required. Combining these inputs, checking them against predefined red flags before any advice, tracking change over time and preparing a summary for a professional must be repeatable and available at every check-in. Manual notes or a coach's memory cannot apply the same safety rules each time. An engineered system can apply them consistently, store history securely and share it only with the user's permission.

BodyBalance AI addresses one question: "Considering how I feel and what is happening with my body today, what is an appropriate next step?" It is a decision-support platform. It combines a physical and mental check-in, a Pain Locator, optional BodyScope AI posture analysis and history to recommend activity, recovery, monitoring or professional evaluation. Diagnosis and treatment remain with qualified professionals.

User scenario

The scenario below is illustrative. It follows the active student persona, a 22-year-old gym user.

1. The student planned a workout. Her shoulder has ached since yesterday and she slept badly.
2. She completes the short daily check-in and marks her shoulder on the body map, rating the pain from 1 to 10.
3. She optionally uploads a posture image. The system checks her answers against the red-flag list before giving any advice.
4. No red flag is found, so the app recommends modifying today's workout, gives a recovery plan and schedules a follow-up.

If a red flag were present, routine guidance would stop and the student would see a warning with referral guidance instead.

2. Market Study and Feasibility

This study reviews the digital health, fitness and recovery market to identify customer needs, competitor strengths, existing gaps and the opportunity for BodyBalance AI. These applications usually specialize: nutrition, workouts, activity tracking, pain management or rehabilitation. BodyBalance AI addresses a different question: "Considering how I feel and what is happening with my body today, what is an appropriate next step?"

2.1 Competitor analysis

Four platforms were selected because they cover the areas closest to BodyBalance AI: nutrition and lifestyle tracking, personalized fitness, pain management and rehabilitation, and wearable health monitoring. The aim is not to show more features than each competitor, but to see what each does well and where a different combination is possible.

Competitor	Main services	Strengths	Limitation / gap
MyFitnessPal	Nutrition tracking, calorie and macronutrient logging, exercise logging, progress tracking	Strong nutrition focus, large food database, established user base	Focused on nutrition and lifestyle, not pain assessment, physical-condition analysis or recovery decisions
Freeletics	AI-based personalized workouts, strength, running, bodyweight training, adaptive plans	Strong workout personalization and fitness-focused AI	Focused on performance and planning, not structured pain, symptom or physical-condition assessment
Hinge Health	Digital musculoskeletal care, pain management, rehabilitation exercises, movement analysis, professional support	Strong pain-management and rehabilitation approach with professional involvement	Specialized care and rehabilitation, not a broader daily fitness and recovery decision process
Google Health / Fitbit	Activity, sleep, stress and health metrics, recovery and readiness information	Broad health ecosystem and strong wearable monitoring	Tracking and metrics, without structured pain mapping, optional visual body analysis or current-condition decision support
BodyBalance AI (proposed)	Physical and mental check-in, Pain Locator, limited BodyScope AI, personalized activity and recovery guidance, progress tracking, safety escalation	Combines current physical, mental, symptom, activity and optional visual information before recommending the next step	New system needs development, technical validation, AI testing, privacy protection and a controlled initial scope

BodyBalance AI should not compete with each platform in its own area. MyFitnessPal stays stronger in nutrition, Freeletics in workout programming, Hinge Health in clinical rehabilitation and Fitbit in continuous wearable monitoring.

2.2 Feature comparison

Legend: ✓ core capability, △ limited, indirect or secondary, ✕ not a core capability.

Feature	MyFitnessPal	Freeletics	Hinge Health	Google Health / Fitbit	BodyBalance AI
Nutrition and calorie tracking	✓	✕	✕	△	✕
Personalized workout planning	△	✓	✓	△	△
Activity and workout tracking	✓	✓	✓	✓	△
Sleep and continuous wearable tracking	✕	✕	✕	✓	✕
Physical and mental self-check-in	✕	△	△	△	✓
Pain location and symptom reporting	✕	✕	✓	✕	✓
Camera and computer-vision analysis	✕	✕	✓	✕	✓
Posture and alignment analysis	✕	✕	△	✕	✓
Personalized recovery guidance	✕	△	✓	△	✓
Recovery and condition progress tracking	△	△	✓	✓	✓

Feature	MyFitnessPal	Freeletics	Hinge Health	Google Health / Fitbit	BodyBalance AI
Clinical care inside the platform	×	×	✓	×	×
Professional referral and escalation guidance	×	×	✓	×	△
Integrated current-condition decision support	×	×	△	△	△

BodyBalance AI follows one process: current physical and mental state + pain and symptoms + recent activity + history + optional BodyScope information → personalized next step. Its advantage is therefore the integration of information relevant to the user's present condition, not the number of features.

2.3 Customer feedback and market gaps

What users value. Personalized guidance that reflects their condition and progress; progress tracking over time; convenient digital access without an appointment for every basic interaction; and awareness of when recovery is more appropriate than training. Each of these maps to a BodyBalance AI function: the check-in, progress tracking, the single mobile platform and recovery-aware recommendations.

Gaps and concerns.

1. Fragmentation: users need several applications for nutrition, workouts, health tracking and rehabilitation.
2. Limited current-condition awareness: general fitness apps plan from long-term goals and may ignore pain, poor sleep, stress or incomplete recovery.
3. AI accuracy and data concerns: incomplete input, poor-quality images and model limits can affect results, so BodyScope AI must stay within a tested scope.
4. Unclear boundary with medical care: the system must act as decision support, and alert the user and recommend professional evaluation when concerning indicators appear.
5. Localisation: many platforms target international markets, leaving room for later Arabic-language and Saudi-market support.
6. Scope control: coaches, clinicians, combat sports, older adults and first aid are long-term options, but attempting them in the first release would raise complexity, validation effort and safety risk.

2.4 Opportunity and differentiation

Personalized workouts, activity tracking, rehabilitation services and camera-based movement analysis already exist, so using AI or offering more features is not a differentiator. BodyBalance AI differentiates through eight points:

1. Current-condition assessment of pain, fatigue, energy, stress, sleep, symptoms and recent activity.
2. Structured pain assessment through an interactive body map and a pain-intensity scale.
3. Limited BodyScope AI: posture-image analysis from pose landmarks, shoulder inclination and pelvic tilt.
4. Personalized next-step recommendation: maintain, modify, reduce intensity, recover, monitor or seek professional evaluation.
5. Follow-up and progress tracking, showing whether pain or recovery is improving, unchanged or worsening.
6. Safety escalation when predefined concerning indicators appear.
7. Professional information summary to help the user communicate clearly; diagnosis and treatment stay with professionals.
8. Controlled MVP scope: account and profile → check-in → Pain Locator → limited BodyScope AI → guidance → progress tracking → escalation → professional summary → privacy and permission controls.

2.5 Feasibility

Feasibility. The preliminary study supports a focused MVP rather than full commercialization.

- **Technical:** depends on whether the recommendation engine and BodyScope AI perform reliably within a limited scope (posture analysis from pose landmarks).
- **Operational:** depends on safe, private interaction for users, coaches and clinicians.
- **Financial:** Year 1 budget of SAR 150,000 (software and AI 55,000; marketing 30,000; cloud 15,000; administration and legal 15,000; contingency 15,000; testing 10,000; professional consultation 10,000). Planning estimates are a Year 1 net result of -SAR 30,000, Year 2 of +SAR 150,000 and Year 3 of +SAR 440,000. These are assumptions to be revalidated after the MVP pilot.

Initial segment: active students, gym users and recreational athletes, reached through campus communities, social media and a freemium model. Combat sports, older adults, first aid and sensitive injury documentation are deferred expansion scenarios.

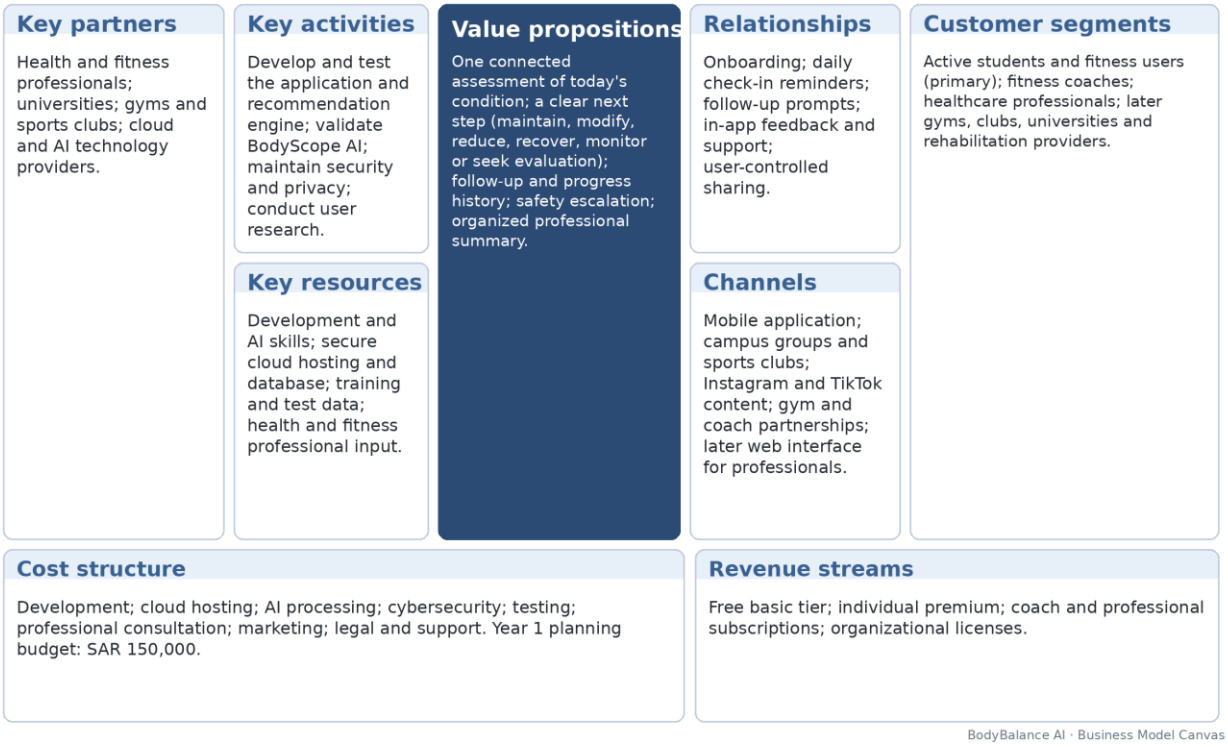
Target area and audience

Segment	Why they need it	How they are reached
Active students, gym users and recreational athletes (first)	They decide daily how hard to train and when to rest	Campus clubs and wellness events, Instagram and TikTok, free tier
Fitness coaches	They adjust client training without a full view of condition	Gym and trainer partnerships
Healthcare professionals	They need organized histories before assessments	Physiotherapy and sports-medicine partners, after validation

The product starts digital-first in Saudi Arabia, with initial testing in university communities, gyms and fitness clubs. Arabic-language support is a later localization step.

3. Business Model Canvas

The model is a freemium mobile application with later professional and organisational plans. Partnerships and willingness to pay are proposals still to be validated in the pilot.



4. Personas

Three personas represent the three user groups. All three were used to derive the use cases and requirements below.

	Active student / fitness user	**Fitness coach**	**Healthcare professional**
Role	Active student and regular gym user, age 22	Fitness coach and personal trainer, age 29	Sports medicine and physiotherapy professional, age 38
Technical level	Medium to high	Medium	High
Goals	Stay active, avoid minor worsening injuries, know when to train or recover, track progress	Train clients more safely, track recovery, adjust exercises to client feedback	Receive organized symptoms and activity history, monitor recovery, identify who needs further evaluation
Pain points	Post-workout pain; unsure whether to rest;	Clients describe pain inaccurately; limited	Patients forget details; histories incomplete;

	Active student / fitness user	Fitness coach	Healthcare professional
	does not know when pain needs a professional; apps ignore her current condition	time; hard to track symptom changes; cannot diagnose	manual review of long-term data is slow
Main features needed	Check-in, Pain Locator, BodyScope AI, activity and recovery guidance, referral summary, progress tracking	Client profile, authorized activity and recovery view, condition alerts, recovery monitoring	Authorized summary, symptoms and activity history, recovery timeline, decision-support information
Usage context	After workouts or when in pain, fatigue or stress	Before or after client sessions, with client permission	Before assessments, through user-authorized records

5. Requirements and Elicitation Methods

5.1 Elicitation methods

Requirements were elicited in layers. Methods already applied are listed first; methods planned before the pilot are marked as planned, since no field interviews or surveys have yet been run.

Method	Status	What it produced
Competitor and document analysis (MyFitnessPal, Freeletics, Hinge Health, Google Health / Fitbit; app store reviews)	Done (market study)	Market gaps, expected features, and the concerns of accuracy, privacy and scope
Persona development	Done (Lab 2)	Three personas and a persona-to-feature map
Scenarios and user stories	Done (Labs 2 and 3)	24 scenarios and the user story set
Story-to-requirement transformation using SMART criteria	Done (Lab 3), revised here	The requirement baseline in 5.2
Use case modelling and UML	Done (Lab 4)	Five use cases, activity and sequence diagrams, domain model

Method	Status	What it produced
Prototype walkthrough (shoulder-pain demo)	Done	Early check on the check-in, Pain Locator and combined summary flow
Team workshops and cross-review	Done	Agreed scope, consistency checks between documents
Interviews with active students (about 8 to 10), coaches and a physiotherapist	Planned	Validation of personas, red-flag list and recommendation categories
Short questionnaire on training and recovery habits	Planned	Evidence of need and willingness to pay
Usability test of the prototype	Planned	Evidence for the usability and performance targets

5.2 User Stories

ID	User story	Persona
US1	As a new user, I want to register, log in securely and keep a profile, so that my records are private and my guidance is personal.	Student
US2	As a user, I want a short daily check-in on fatigue, energy, stress, sleep, activity and symptoms, so that the system knows how I feel today.	Student
US3	As a user, I want to mark where it hurts on a body map and rate the pain, so that my pain is described precisely.	Student
US4	As a user, I want to upload a posture image and get simple posture measurements, so that I can see my alignment.	Student
US5	As a user, I want one clear recommendation with reasons, so that I know whether to train, modify or rest.	Student
US6	As a user, I want a recovery plan that fits my pain areas and past injuries, so that I recover safely.	Student
US7	As a user, I want a timeline and charts of my pain and posture, so that I can see my progress.	Student
US8	As a user, I want follow-up prompts that update my next recommendation, so that guidance follows my recovery.	Student

ID	User story	Persona
US9	As a user, I want a PDF summary of my condition, so that I can explain it clearly to a professional.	Student
US10	As a user, I want serious symptoms to stop routine advice and tell me to seek care, so that I am not misled.	Student
US11	As a coach, I want to see the activity and recovery data my client allows and be alerted to high pain or fatigue, so that I adjust training safely.	Coach
US12	As a healthcare professional, I want to view the history my patient allows, so that I assess faster and with full information.	Professional
US13	As a user, I want to grant and revoke access per professional and data category, so that I control my data.	Student
US14	As a physiotherapist, I want to review and change AI-generated exercise plans before they reach my patient, so that the plan is clinically safe.	Professional

5.3 Final functional requirements (SMART)

The final requirements are the 34 below. Each has one ID, a user story and a pass or fails acceptance measure, so it can be tested. Numeric thresholds are initial targets to be tuned during testing.

ID	**Story**	**Requirement**	**Acceptance measure**
REQ-1	US1	The system shall let a new user register with name, age, email and password after accepting the privacy and consent notice.	Account created within 3 s for 100% of valid submissions; registration blocked when consent is not accepted.
REQ-2	US1	The system shall authenticate users by email and password (minimum 8 characters), allow password reset by a link valid for 30 minutes, and end a session after 30 minutes of inactivity.	Test cases for login, reset, expiry and timeout all pass; account locked for 15 minutes after 5 failed attempts.
REQ-3	US1	The system shall let a user create and edit a profile with age, activity level, fitness background, goals, physical limitations and previous injuries.	Profile changes saved within 2 s and used in the next recommendation.
REQ-4	US2	The system shall let the user complete a daily check-in covering fatigue, energy and stress (each rated 1 to 10),	Median completion time of 3 minutes or less in usability testing.

ID	Story	Requirement	Acceptance measure
		sleep hours, recent activity and symptoms.	
REQ-5	US2	The system shall reject a check-in with missing required fields and name each missing field.	100% of stored check-ins contain all required fields.
REQ-6	US2	The system shall save each check-in with date and time on submission.	Saved within 2 s; retrievable in history.
REQ-7	US3	The system shall provide an interactive body map for selecting up to 3 distinct pain zones per assessment.	A fourth selection is blocked with a message.
REQ-8	US3	The system shall record, for each selected zone, a pain intensity from 1 to 10, onset date, cause (activity, injury or unknown) and change (better, same, worse).	No zone can be saved without an intensity.
REQ-9	US3	The system shall store the selected zones and their details in the assessment record on submission.	Stored values equal submitted values in 100% of test cases.
REQ-10	US4	The system shall accept posture images in JPEG or PNG format up to 10 MB and reject other files.	Invalid files rejected with a message within 1 s.
REQ-11	US4	The system shall check lighting and full-body visibility of an uploaded image within 3 s.	Image passes only if brightness is within a set range and at least 30 of 33 landmarks are detected at confidence 0.5 or higher; otherwise a guided retake prompt is shown.
REQ-12	US4	BodyScope AI shall extract 33 pose landmarks and calculate shoulder inclination and pelvic tilt in degrees from each valid image within 5 s.	At least 95% of valid images processed within 5 s.
REQ-13	US4	BodyScope AI shall report posture measurements as non-diagnostic observations.	Shoulder and pelvic values within 3 degrees of a manual reference for at least 90% of a 30-image test set; label shown on 100% of results.
REQ-14	US5	The system shall generate a recommendation from the profile,	Delivered within 5 s for at least 95% of requests; test case without

ID	Story	Requirement	Acceptance measure
		check-in, pain zones and intensity, recent activity and available BodyScope results, and shall not invent posture results when BodyScope is unavailable.	an image produces no posture values.
REQ-15	US5	The system shall return exactly one of six outcomes: maintain activity, modify activity, reduce intensity, focus on recovery, monitor, or consider professional evaluation.	100% of output fall in one category.
REQ-16	US5	The system shall display an explanation naming at least the three most influential user-reported factors (all, if fewer than three were given).	Explanation present for 100% of recommendations.
REQ-17	US6	The system shall generate a recovery plan with rest guidance, mobility guidance and activity restrictions, and set a follow-up date within 72 hours.	All three components and a follow-up date are present in 100% of plans.
REQ-18	US6	The recovery plan shall address each selected pain zone.	Every selected zone appears in the plan in test cases.
REQ-19	US6	The system shall not recommend exercises that load a body area recorded as a limitation or previous injury.	0 conflicts across the test set.
REQ-20	US7	The system shall show check-ins, pain records, posture results and recovery plans in chronological order in a Body Timeline.	Timeline loads within 3 s for 90 days of data.
REQ-21	US7	The system shall chart pain intensity and posture measurements per zone over 7, 30 or 90 days.	Chart matches stored values in 100% of test cases.
REQ-22	US8	The system shall prompt for follow-up on the follow-up date set in the recovery plan and record pain intensity (1 to 10) and status (improved, unchanged, worse).	Prompt delivered on schedule in 100% of test cases.

ID	Story	Requirement	Acceptance measure
REQ-23	US8	The system shall use the latest follow-up in the next recommendation and shall recommend professional evaluation after two consecutive "worse" follow-ups.	Verified by scripted test cases; threshold to be confirmed by a clinical advisor.
REQ-24	US9	The system shall generate a downloadable PDF referral summary of symptoms, affected areas, recent activity, progress and BodyScope observations for a user-selected period.	Generated within 10 s.
REQ-25	US9	The summary shall state that it is decision-support information, contains no diagnosis, and includes only categories the user selects.	Label present in 100% of summaries; no unselected category appears.
REQ-26	US10	The system shall check every submission against the eight predefined red-flag indicators (Section 5.3) before generating any recommendation.	All eight evaluated 100% of submissions.
REQ-27	US10	When a red flag is detected, the system shall block routine recommendation generation and display a safety warning with referral guidance (urgent-care wording for chest pain, breathing difficulty or fainting).	Warning displayed within 2 s; 0 routine plans generated in red-flag test cases.
REQ-28	US11	The system shall let an authorized coach view only the fitness, activity and recovery categories the client has permitted.	100% of requests outside the permitted scope denied.
REQ-29	US11	The system shall notify an authorized coach when a consenting client records pain of 7 or more or fatigue of 8 or more.	Notification within 5 minutes of submission.
REQ-30	US12	The system shall let an authorized healthcare professional view the permitted symptom history, pain	Record opens within 3 s.

ID	Story	Requirement	Acceptance measure
		records, posture assessments and recovery progress for the permitted period.	
REQ-31	US12	The system shall grant professional access only after the professional's identity is verified.	0 successful access by unverified accounts.
REQ-32	US13	The system shall let a user grant and revoke access per professional and per data category.	Changes stored and effective within 5 s.
REQ-33	US13	The system shall deny access when permission is missing, revoked or expired and show that no authorized information is available.	100% of requests after revocation were denied from 5 s after the change.
REQ-34	US14	For a user with a linked physiotherapist, the system shall hold the AI-generated exercise plan as pending review and publish it only after the physiotherapist approves, modifies, replaces or disables exercises. Red-flag warnings are never held.	Edits logged with date and time; no plan published before review.

5.4 Red-flag indicators (provisional)

The system checks eight predefined red-flag indicators before giving any routine activity guidance. The list below makes REQ-26 and REQ-27 testable. It is provisional and must be reviewed and approved by a healthcare advisor before the pilot.

#	Indicator reported by the user
1	Chest pain or pressure
2	Difficulty breathing at rest
3	Fainting or severe dizziness
4	Pain intensity of 8 or more out of 10
5	New numbness, tingling or muscle weakness
6	Visible deformity, or inability to move or bear weight after an injury
7	Head injury with confusion, repeated vomiting or severe headache

#	Indicator reported by the user
8	Rapidly increasing swelling, or fever with joint pain or redness

5.5 Non-functional requirements (measurable)

The non-functional requirements below are stated with measurable targets, so each can be verified by a test.

ID	Category	Requirement	Acceptance measure
NFR-1	Security	All protected data shall require authentication, be encrypted in transit (TLS 1.2 or higher) and at rest (AES-256), and every professional request shall be checked against active permissions.	100% of unauthenticated or unpermitted requests are denied in security tests.
NFR-2	Performance	The system shall show a processing status (pending, completed or failed) within 1 s of any assessment, recommendation or image submission.	Status visible within 1 s in 100% of test runs.
NFR-3	Usability	A new user shall complete registration, profile and a first check-in unaided.	At least 80% do so in 10 minutes or less; System Usability Scale score of 70 or higher in the pilot.
NFR-4	Reliability	The system shall mark a submission as completed only when all required data is stored, and shall be available during normal operation.	0 submissions marked complete without stored data; 99% monthly availability.
NFR-5	Safety	All safety checks shall finish before any routine recommendation is shown.	100% of red-flag test cases produce the escalation response.
NFR-6	Transparency	Every BodyScope, activity, recovery and referral output shall be labelled as decision-support information, not a diagnosis.	Label present on 100% of outputs.
NFR-7	Localization (future)	The interface shall support Arabic and English.	Deferred beyond the MVP.

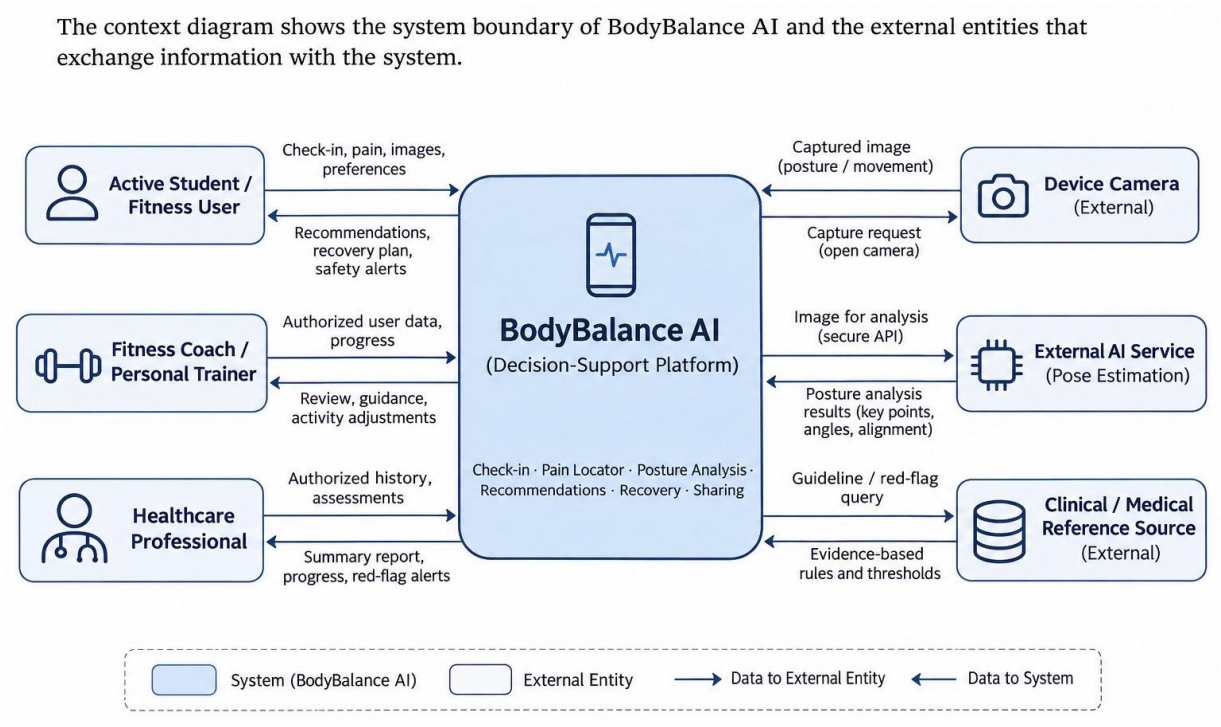
5.6 SMART check

Criterion	Result
Specific	Each requirement names one actor, one function and the data involved.
Measurable	Each has a numeric or pass or fail acceptance measure. Initial thresholds are marked for tuning.
Achievable	Limited to the MVP scope: posture analysis from pose landmarks, six recommendation outcomes, eight red flags.
Relevant	Each trace to a user story, use case and persona (Section 6).
Time-bound	Response, delivery and revocation times are stated in seconds, minutes or hours.

6. Context Diagram

The context model fixes the system boundary and shows the people, device inputs and external services that exchange information with BodyBalance AI. The user remains the primary actor; the device camera or photo library supplies posture images; the external pose-estimation/AI service returns posture observations; and the clinical/medical reference source provides the safety criteria used by the decision-support workflow. Coaches and healthcare professionals receive only user-authorised information.

The context diagram shows the system boundary of BodyBalance AI and the external entities that exchange information with the system.



context diagram · BodyBalance AI system boundary and external entities

The user supplies check-in data, pain zones, posture images, follow-up feedback and sharing permissions, and receives recommendations, recovery plans, safety warnings, progress views and referral summaries. Coaches and healthcare professionals receive only the information the user has authorized, so every flow to them depends on a user permission (REQ-28 to REQ-33). The physiotherapist can also review and change AI-recommended exercises before publication (REQ-34).

Communication view (UC-03)

The main scenario of UC-03 involves six objects: the user, the mobile app, the risk and escalation logic, the recommendation engine, the database and BodyScope AI. Messages occur in this order:

1. The user submits the assessment to the app.
2. The app asks the risk and escalation logic to check red flags.
3. The app retrieves the profile, check-ins, pain records and history from the database. BodyScope AI results, when available, were stored there during UC-02.
4. The app asks the recommendation engine to generate guidance.
5. The app shows the recommendation and explanation to the user.

If message 2 finds a red flag, the app shows the safety warning and referral guidance and skips message 4. Object names follow the system modules in the feasibility study and may change in the final design.

Operational aspects. Running the service requires cloud hosting and backups, security updates, model evaluation and maintenance, and user support. These appear in the cost structure (Section 3) and the Year 1 budget (Section 2).

7. Use Cases

The system's functions are grouped into five goal-level use cases. Each has a use case diagram, an activity diagram and a sequence diagram. The table below lists the use cases with the functions they cover and the requirements that trace to each, including data sharing and physiotherapist review.

ID	Use case	Primary actor	Priority	Merged functions	Requirements
UC-01	Manage User Account	Active Student / Fitness User	Must	Register, authenticate, manage profile	REQ-1 to REQ-3
UC-02	Perform Body Assessment	Active Student / Fitness User	Must	Daily check-in, pain assessment, posture analysis	REQ-4 to REQ-13
UC-03	Generate Safe Activity Recommendation	Active Student / Fitness User	Must	Safety check, activity recommendation, conditional referral guidance	REQ-14 to REQ-16, REQ-26, REQ-27

ID	Use case	Primary actor	Priority	Merged functions	Requirements
UC-04	Manage Recovery and Progress	Active Student / Fitness User	Must	Recovery plan, follow-up feedback, progress history	REQ-17 to REQ-23
UC-05	Manage Professional Access	Active Student / Fitness User (coach and healthcare professional as supporting actors)	Should	Permissions, professional summary, coach review, healthcare review, physiotherapist plan review	REQ-24, REQ-25, REQ-28 to REQ-34

UC-03 is the most complex case and is fully specified in Lab 4: its main flow is submitting assessments, validate, safety check, retrieve data, generate recommendation, explain. Its alternatives are missing information, a concerning indicator (block routine guidance and refer) and BodyScope unavailable (continue without posture results). UC-05 is specified as the authorization edge case: access is denied when permission is missing or expired, and professionals never receive information outside the user-authorized scope.

7.1 Use Case Specifications and Diagrams

Each goal-level use case is specified below and followed by its diagram. Include arrows point from the base use case to the included one; extend arrows point from the extending use case to its base.

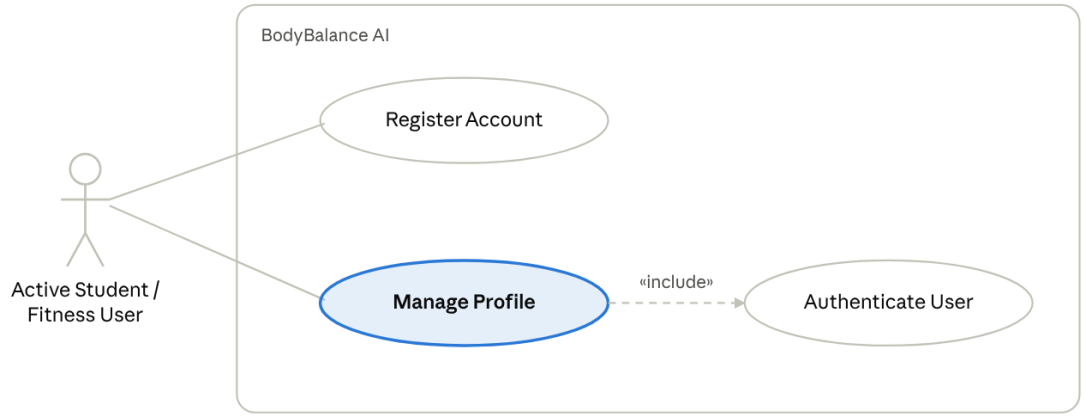
UC-01 Manage User Account

Field	Content
Primary actor	Active student / fitness user
Goal	Create, access and maintain an account
Preconditions	BodyBalance AI is available
Postconditions	Account and profile changes are securely stored
Main success scenario	The user registers an account or manages profile information; authentication is required before protected profile management

Traceability for UC-01

Use Case	User Stories	Requirements
UC-01	US1	REQ-1 to REQ-3

UC-01: profile management includes authentication



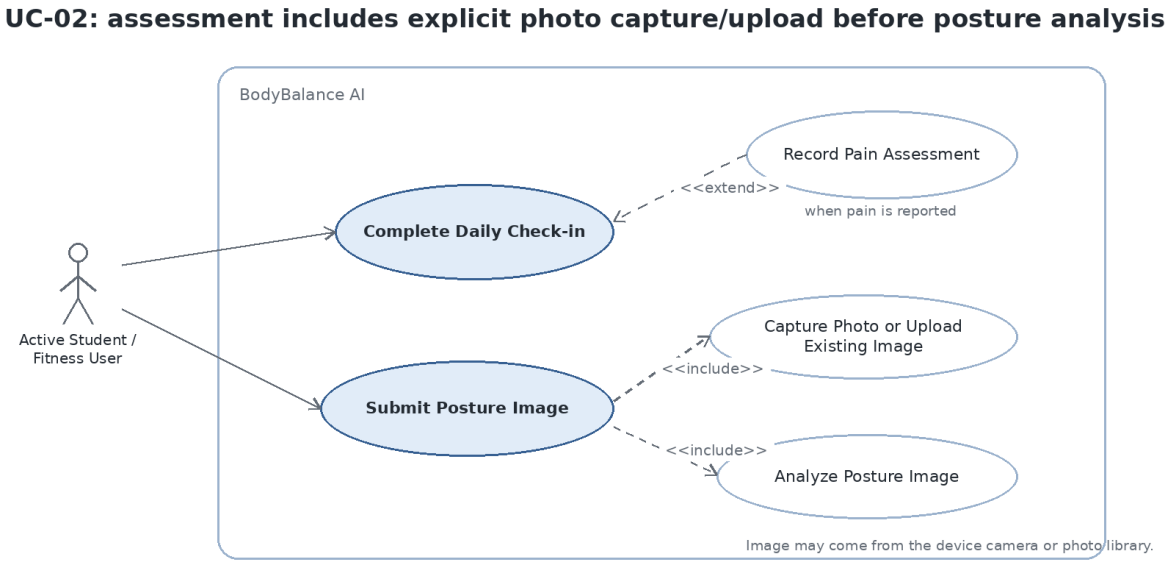
UC-01 use case diagram

UC-02 Perform Body Assessment

Field	Content
Primary actor	Active student / fitness user
Goal	Capture the user's current condition
Preconditions	The user is authenticated
Postconditions	Current assessment information is validated and stored
Main success scenario	The user completes the daily check-in. Pain assessment is added when pain is reported. For posture assessment, the user captures a photo with the device camera or uploads an existing JPEG/PNG image; the system validates the image and then sends it for BodyScope AI posture analysis.

Traceability for UC-02

Use Case	User Stories	Requirements
UC-02	US2, US3, US4	REQ-4 to REQ-13



UC-02 use case diagram

UC-03 Generate Safe Activity Recommendation

Field	Content
Primary actor	Active student / fitness user
Stakeholders and interests	User: safe, relevant guidance. Fitness coach: recommendations that reflect relevant client information. Healthcare professional: AI output that stays decision-support information
Preconditions	The user is authenticated; current assessment information is complete enough; applicable profile and assessment information can be retrieved
Postconditions	A personalised recommendation is displayed with an explanation. If a concerning indicator is detected, routine exercise guidance is blocked and referral guidance is provided
Special requirements	The main user-reported factors behind the recommendation are identified. AI output is shown as decision-support information, not a confirmed diagnosis
Constraints	Diagnosis and treatment are outside the system's role
Open issues	Exact recommendation logic and supported activity categories need later validation and testing

Traceability for UC-03

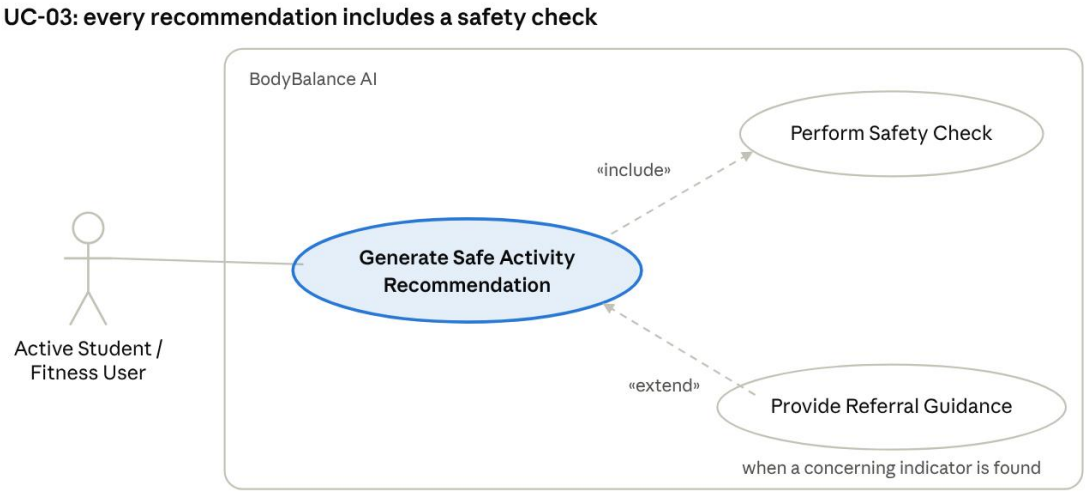
Use Case	User Stories	Requirements
UC-03	US5, US10	REQ-14 to REQ-16; REQ-26 to REQ-27

Main success scenario:

1. The user submits the current assessment.
2. The system validates the required information.
3. The system performs a safety check.
4. The system retrieves the applicable profile, check-in, pain, activity and available BodyScope information.
5. The recommendation engine generates activity guidance.
6. The system generates and displays the recommendation explanation.
7. The user reviews the recommendation.

Alternatives:

- 2a. Missing information: request completion before continuing.
- 3a. Concerning indicator: block routine exercise guidance, display a warning and provide referral guidance.
- 4a. BodyScope unavailable: use the other available information without inventing posture results.



UC-03 use case diagram

UC-04 Manage Recovery and Progress

Field	Content
Primary actor	Active student / fitness user
Goal	Support recovery and progress over time
Preconditions	Relevant assessment or recovery information exists
Postconditions	Recovery guidance, follow-up feedback and progress history are available
Main success scenario	The system generates recovery guidance; the user later submits follow-up feedback and reviews progress over time

Traceability for UC-04

Use Case	User Stories	Requirements
UC-04	US6, US7, US8	REQ-17 to REQ-23



UC-04 use case diagram

UC-05 Manage Professional Access

Field	Content
Primary actor	Active student / fitness user
Supporting actors	Fitness coach / personal trainer; healthcare professional
Stakeholders and interests	User: controls what is shared. Fitness coach: relevant fitness, activity and recovery information. Healthcare professional: relevant symptom, activity, posture and recovery information

Field	Content
Preconditions	The user is authenticated; the professional's identity is valid; access is based on user authorisation
Postconditions	Permissions are stored; only authorised information is available to the selected professional; an organised summary may be generated for sharing
Special requirements	Professional access remains user-controlled and privacy-preserving
Constraints	Professionals must not receive information outside the user-authorised scope
Open issues	Exact authorisation duration and revocation workflow need implementation validation

Traceability for UC-05

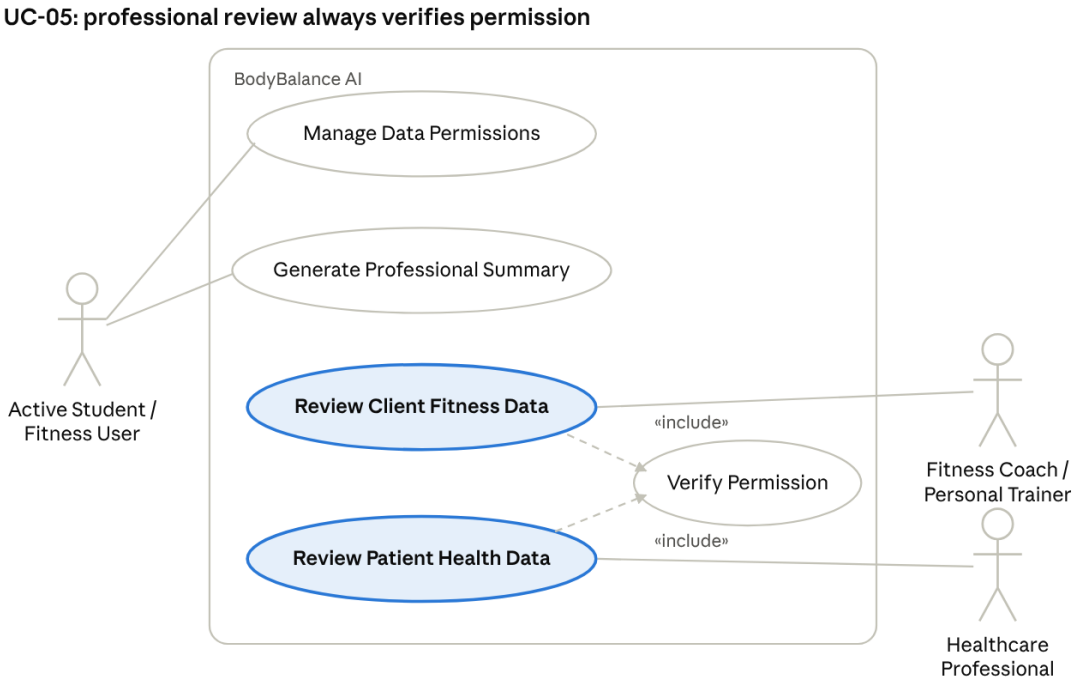
Use Case	User Stories	Requirements
UC-05	US9, US11, US12, US13, US14	REQ-24 to REQ-25; REQ-28 to REQ-34

Main success scenario:

1. The user opens the professional-access settings.
2. The user selects the professional and the information to share.
3. The system validates and stores the permissions.
4. The user may generate a professional summary.
5. The authorised professional opens the permitted record.
6. The system displays only the authorised information.

Alternatives:

- 3a. Authorisation missing or expired: access is denied.
- 5a. No permitted records: display that no authorised information is available.



UC-05 use case diagram

8. System Architecture

The BodyBalance AI system architecture was designed based on the project requirements and findings from the literature review. The architecture follows a modular approach in which data collection, physical assessment, safety screening, recommendation generation and user feedback are separated into connected components. This separation improves maintainability, testability and safety while preventing a single AI component from controlling the complete decision-making process.

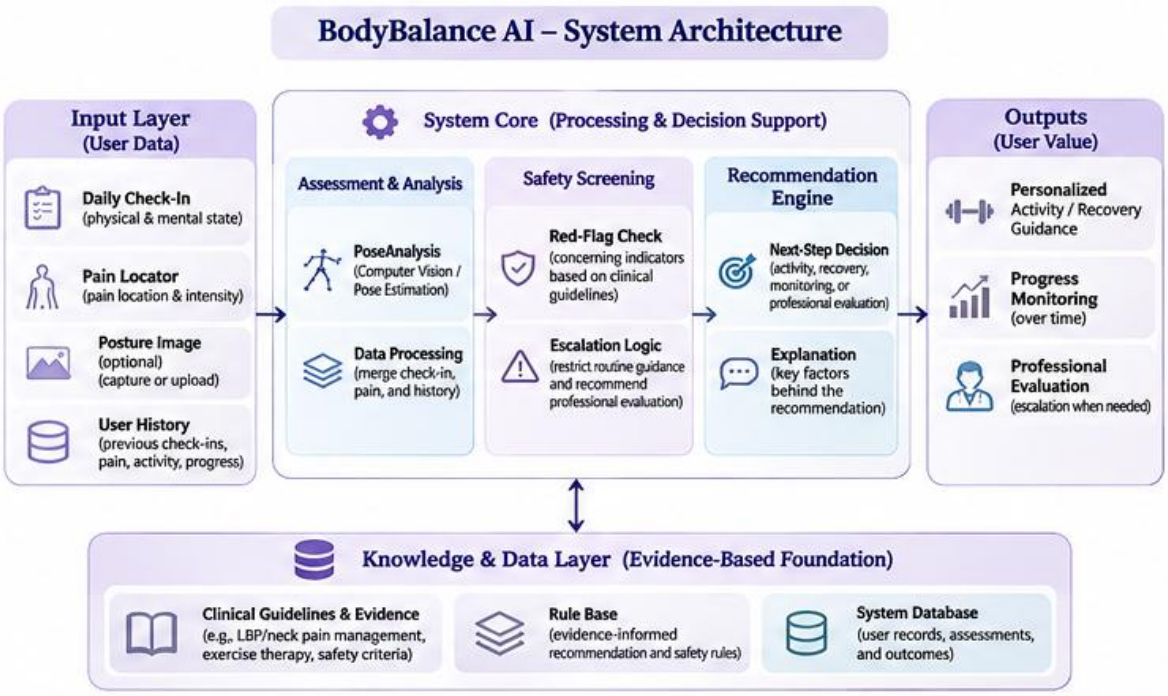


Figure 8.1. Proposed System Architecture of BodyBalance AI based on the literature review and system requirements.

User and Input Layer. The architecture begins with the user providing multimodal information through the daily physical and mental check-in, Pain Locator, recent activity and user history, and an optional posture image. The Pain Locator records structured pain location and intensity, while BodyScope receives a captured or uploaded posture image when the user chooses to use this feature.

Processing and Assessment Layer. Structured check-in and pain information are combined with relevant historical data. The optional BodyScope component applies computer-vision and pose-estimation techniques to identify body landmarks and derive posture-related features such as alignment or asymmetry. These features are treated as supporting information and are not used as a medical diagnosis.

Safety Screening Layer. Before routine guidance is produced, the collected information passes through a separate safety-screening component. Predefined concerning indicators are checked before the recommendation stage. If a safety condition is triggered, routine guidance is restricted and the system directs the user toward professional evaluation. The safety component is a screening and escalation mechanism rather than a diagnostic model.

Recommendation and Explanation Layer. When no safety restriction is triggered, the recommendation component combines the current check-in, pain information, historical context and optional BodyScope features to determine an appropriate next-step category: activity, recovery, continued monitoring or professional evaluation. The result is accompanied by a short explanation of the main factors that influenced the recommendation.

Knowledge and Data Layer. The architecture is supported by system data and evidence-informed rules. User records, previous assessments and recommendation history provide temporal context, while relevant clinical literature and guidelines inform the boundaries of safety and recommendation rules. The final rules require professional review and validation before any clinical deployment.

Professional Escalation and System Boundary. When escalation is appropriate, the system can prepare relevant information for professional review according to the user's sharing permissions. AI therefore operates as a supporting capability within a controlled software workflow rather than as an autonomous medical decision-maker. This architecture reflects the literature-review findings on multimodal assessment, explicit safety boundaries, explainability and human oversight.

9. Project Management

Approach. An iterative, MVP-first plan. The first release covers one user journey: account and profile, check-in, Pain Locator, limited BodyScope AI, recommendation, recovery, follow-up, escalation and professional summary. Specialized scenarios (combat sports, older adults, first aid, sensitive injury documentation) follow after testing, professional review and privacy assessment.

Team and roles (equal 33.3% ownership proposed):

Member	Responsibility
Leen Kasem	Product development, software development, AI, project management
Reema Abualnaja	Product development, software development, AI, project management
Farah Shetawi	Marketing, documentation, administrative support

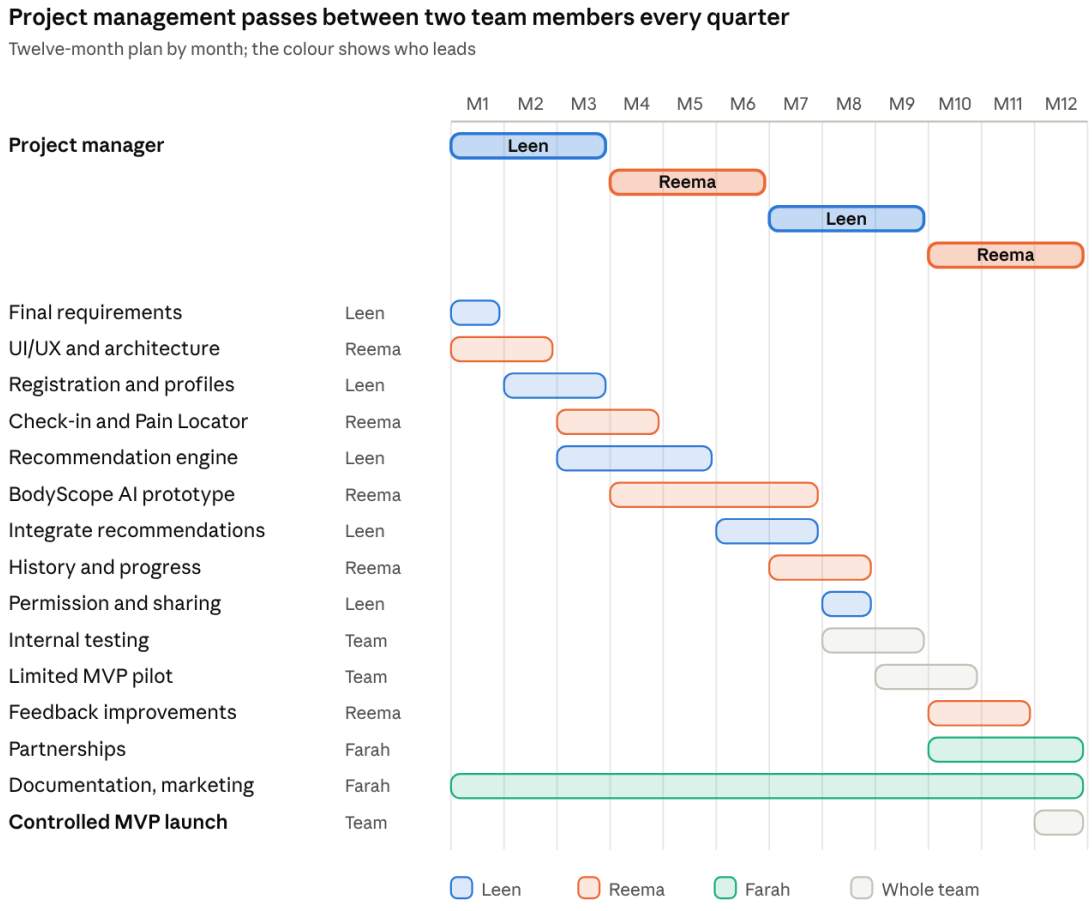
The team starts with 3 to 4 people, grows to 5 to 6 within six months (AI and computer vision, cybersecurity, part-time health and fitness advisers) and to 7 to 10 in the first year, depending on MVP results and funding.

Twelve-month plan

Months	Activity
1	Finalize business and software requirements
1 to 2	UI/UX prototype and technical architecture
2 to 3	Registration, authentication and profiles
3 to 4	Check-in and Pain Locator
3 to 5	Initial recommendation engine
4 to 7	BodyScope AI prototype
6 to 7	Integrate activity and recovery recommendations
7 to 8	History, follow-up and progress tracking
8	Permission and information sharing

Months	Activity
8 to 9	Internal testing and technical evaluation
9 to 10	Limited MVP pilot
10 to 11	Improvements from feedback
10 to 12	Fitness and health partnerships
12	Controlled MVP launch

Gantt chart and project manager rotation. The chart places the twelve-month plan on a monthly scale and shows who leads each activity. Leen and Reema alternate as project managers every quarter, so each manages two quarters; the project manager owns the plan, risks and status reporting, while the other keeps leading her own activities. Farah leads documentation and marketing throughout and partnerships in months 10 to 12.



Gantt chart · 12 months, project manager rotation

This rotation and the activity leads are the team's proposed plan, based on the members responsibilities above, and are to be confirmed with the instructor.

Working practices. The team discusses decisions together, cross-reviews requirements and models, and keeps documents consistent through one requirement ID scheme. Image upload also needs quality checks, secure storage and permissions.

Key risks and responses.

Risk	Response
Inaccurate recommendations	Conservative logic, testing and red-flag escalation
BodyScope errors or poor images	Image-quality checks, limited supported uses, non-diagnostic wording
Privacy breach or unauthorized sharing	Encryption, access control, minimum data, revocable permissions
User over-reliance on AI	Clear limitations and professional referral prompts
Scope too broad	Focused MVP and gradual expansion
High AI and cloud cost	Model optimization and scalable infrastructure
Regulatory change	Early regulatory review and adaptable scope

Budget. Year 1 planning budget of SAR 150,000 (Section 2). Estimated revenue is SAR 120,000 in Year 1, SAR 400,000 in Year 2 and SAR 850,000 in Year 3; these are assumptions to be recalculated after the pilot.

10. Traceability Matrix

Each user story maps to the requirements that realise it and to the use case that contains it, so every requirement can be traced to a user need and a goal-level use case.

Story	User story	Requirements	Use case
US1	Registration and profile	REQ-1 to REQ-3	UC-01
US2	Mental and physical check-in	REQ-4 to REQ-6	UC-02
US3	Pain locator	REQ-7 to REQ-9	UC-02
US4	BodyScope AI posture analysis	REQ-10 to REQ-13	UC-02
US5	Activity recommendation	REQ-14 to REQ-16	UC-03
US6	Recovery plan	REQ-17 to REQ-19	UC-04
US7	Body timeline	REQ-20, REQ-21	UC-04

Story	User story	Requirements	Use case
US8	Follow-up feedback	REQ-22, REQ-23	UC-04
US9	Professional referral	REQ-24, REQ-25	UC-05
US10	Safety escalation	REQ-26, REQ-27	UC-03
US11	Coach access	REQ-28, REQ-29	UC-05
US12	Healthcare professional access	REQ-30, REQ-31	UC-05
US13	Permissions and professional summary	REQ-32, REQ-33	UC-05
US14	Physiotherapist plan review	REQ-34	UC-05

11. SDGs and Saudi Vision 2030

BodyBalance AI contributes most directly to health and well-being, and supports innovation and partnerships through its delivery model.

Framework	**How the project aligns**	**Linked requirements**
UN SDG 3: Good Health and Well-being	Helps active people judge when to train, recover or seek care; escalates concerning symptoms to professional evaluation; tracks recovery and mental state (stress, sleep, energy)	REQ-4 to REQ-9, REQ-14 to REQ-23, REQ-26, REQ-27
UN SDG 9: Industry, Innovation and Infrastructure	Applies AI and computer vision to a digital health service built by a local team on scalable cloud infrastructure	REQ-10 to REQ-14, NFR-1 to NFR-6
UN SDG 17: Partnerships for the Goals	Works with universities, gyms, clubs and healthcare professionals for testing, referral and validation	REQ-28 to REQ-34
UN SDG 10: Reduced Inequalities (future)	Arabic-language support and low-cost freemium access can widen reach	NFR-7
Saudi Vision 2030: Quality of Life Program	Encourages regular, safer physical activity and healthier lifestyles, starting with university students and gym users	REQ-14 to REQ-19

Framework	How the project aligns	Linked requirements
Saudi Vision 2030: Health Sector Transformation	Digital, preventive and data-supported health guidance, with clear referral to professional care	REQ-24 to REQ-27, REQ-30
Saudi Vision 2030: Digital economy, entrepreneurship and data and AI strategy	A student-founded AI product with a freemium and subscription model; privacy and consent controls prepare for Saudi data-protection requirements (to be confirmed before launch)	REQ-1 to REQ-3, REQ-32, REQ-33, NFR-1

The project stays within decision support. Its contribution to these goals depends on the pilot confirming that recommendations are safe, accurate within their limited scope, and trusted by users and professionals.

Conclusion

The analysis phase gives BodyBalance AI a consistent baseline: a clearly stated problem, a market gap between goal-based fitness applications and clinical services, a freemium business model, three personas, functional and non-functional requirements that can each be tested, five use cases and a defined system boundary. The platform's value lies in integrating a person's current physical and mental state, pain, activity, history and an optional image into one next-step recommendation, with safety escalation and user-controlled sharing.

The main open items are validation steps rather than design gaps. The red-flag list and recommendation thresholds need review by a healthcare adviser, and user interviews, a usability test and technical testing of BodyScope AI are still to be done. Applicable Saudi privacy and digital-health requirements must be confirmed before launch. The recommended next step is a focused MVP and pilot, not full commercialization.

References

Elkafrawy, P. (2026a). *Software engineering lab: From personas and requirements to use cases and UML* [Course handout]. CS3151 Software Engineering, Effat University.

Elkafrawy, P. (2026b). *Chapter 4: System modeling* [Lecture material, based on SWEBOK v4, Chapter 3]. CS3151 Software Engineering, Effat University.

IEEE Computer Society. (2024). *Guide to the software engineering body of knowledge (SWEBOK Guide)* (Version 4.0).

BodyBalance AI project team. (2026). *Labs 1 to 4 and feasibility study* [Unpublished course documents]. CS3151 Software Engineering, Effat University.

Competitor information in Section 2 draws on the official websites and support pages of MyFitnessPal, Freeletics, Hinge Health and Google/Fitbit, and on app store reviews, as listed in the Lab 1 market study.